

1. T1 — Monthly Performance

```
SELECT
    EXTRACT(YEAR FROM o.order_date) AS year,
    EXTRACT(MONTH FROM o.order_date) AS month,
    COUNT(DISTINCT o.order_id) AS total_orders,
    SUM(od.unit_price * od.quantity * (1 - od.discount)) AS total_revenue,
    AVG(od.unit_price * od.quantity * (1 - od.discount)) AS avg_order_value
FROM orders o
JOIN order_details od ON o.order_id = od.order_id
WHERE EXTRACT(YEAR FROM o.order_date) = 2025
GROUP BY year, month
ORDER BY month;
```

2. T2 — Shipper SLA (YTD)

```
SELECT
    s.company_name AS shipper,
    COUNT(o.order_id) AS shipped_orders,
    COUNT(*) FILTER (WHERE o.shipped_date > o.required_date) AS late_orders,
    ROUND(
        COUNT(*) FILTER (WHERE o.shipped_date > o.required_date)::numeric /
        COUNT(o.order_id)::numeric * 100, 1
    ) AS late_rate_percent
FROM orders o
JOIN shippers s ON o.ship_via = s.shipper_id
WHERE EXTRACT(YEAR FROM o.shipped_date) = 2025
GROUP BY s.company_name
ORDER BY late_rate_percent DESC;
```

3. T3 — Top 10 Products by Revenue (YTD)

```
SELECT
    p.product_name,
    c.category_name,
    SUM(od.unit_price * od.quantity * (1 - od.discount)) AS revenue,
    CASE
        WHEN p.unit_price < 10 THEN '<10'
        WHEN p.unit_price BETWEEN 10 AND 20 THEN '10-20'
        WHEN p.unit_price BETWEEN 20 AND 50 THEN '20-50'
        ELSE '>50'
    END AS price_range
FROM orders o
JOIN order_details od ON o.order_id = od.order_id
JOIN products p ON p.product_id = od.product_id
JOIN categories c ON c.category_id = p.category_id
WHERE EXTRACT(YEAR FROM o.order_date) = 2025
GROUP BY p.product_name, c.category_name, price_range
ORDER BY revenue DESC
LIMIT 10;
```

4. T4 — Sales Rep Scorecard (YTD)

```
WITH order_totals AS (
    SELECT
        o.order_id,
        o.employee_id,
        SUM(od.unit_price * od.quantity * (1 - od.discount)) AS order_total
    FROM orders o
    JOIN order_details od ON o.order_id = od.order_id
```

```

WHERE EXTRACT(YEAR FROM o.order_date) = 2025
GROUP BY o.order_id, o.employee_id
)
SELECT
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name,
    COUNT(ot.order_id) AS total_orders,
    AVG(ot.order_total) AS avg_order_value,
    ROUND(
        COUNT(*) FILTER (WHERE ot.order_total > 1000)::numeric /
        COUNT(ot.order_id)::numeric * 100, 1
    ) AS percent_large_orders
FROM order_totals ot
JOIN employees e ON ot.employee_id = e.employee_id
GROUP BY employee_name
ORDER BY percent_large_orders DESC;

```

5. T5 — New vs Loyal Customers (YTD)

```

WITH customer_first AS (
    SELECT
        customer_id,
        MIN(order_date) AS first_order_date
    FROM orders
    GROUP BY customer_id
),
ytd_orders AS (
    SELECT
        o.order_id,
        o.customer_id,
        SUM(od.unit_price * od.quantity * (1 - od.discount)) AS revenue

```

```
FROM orders o
JOIN order_details od ON o.order_id = od.order_id
WHERE EXTRACT(YEAR FROM o.order_date) = 2025
GROUP BY o.order_id, o.customer_id
)
SELECT
CASE
    WHEN EXTRACT(YEAR FROM cf.first_order_date) = 2025 THEN 'NEW'
    ELSE 'LOYAL'
END AS segment,
COUNT(y.order_id) AS orders,
SUM(y.revenue) AS total_revenue
FROM ytd_orders y
JOIN customer_first cf ON y.customer_id = cf.customer_id
GROUP BY segment
ORDER BY segment;
```